Installing the Jetson Environment

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Some users may need to use the NVIDIA built-in system component environment. After the system is installed and the system desktop is successfully entered, the component environment can be installed according to the tutorial below!

Note: The SSD that comes with the product contains a factory image, and the factory image includes the NVIDIA official system and the mainboard environment; if you do not require a clean system, do not operate!

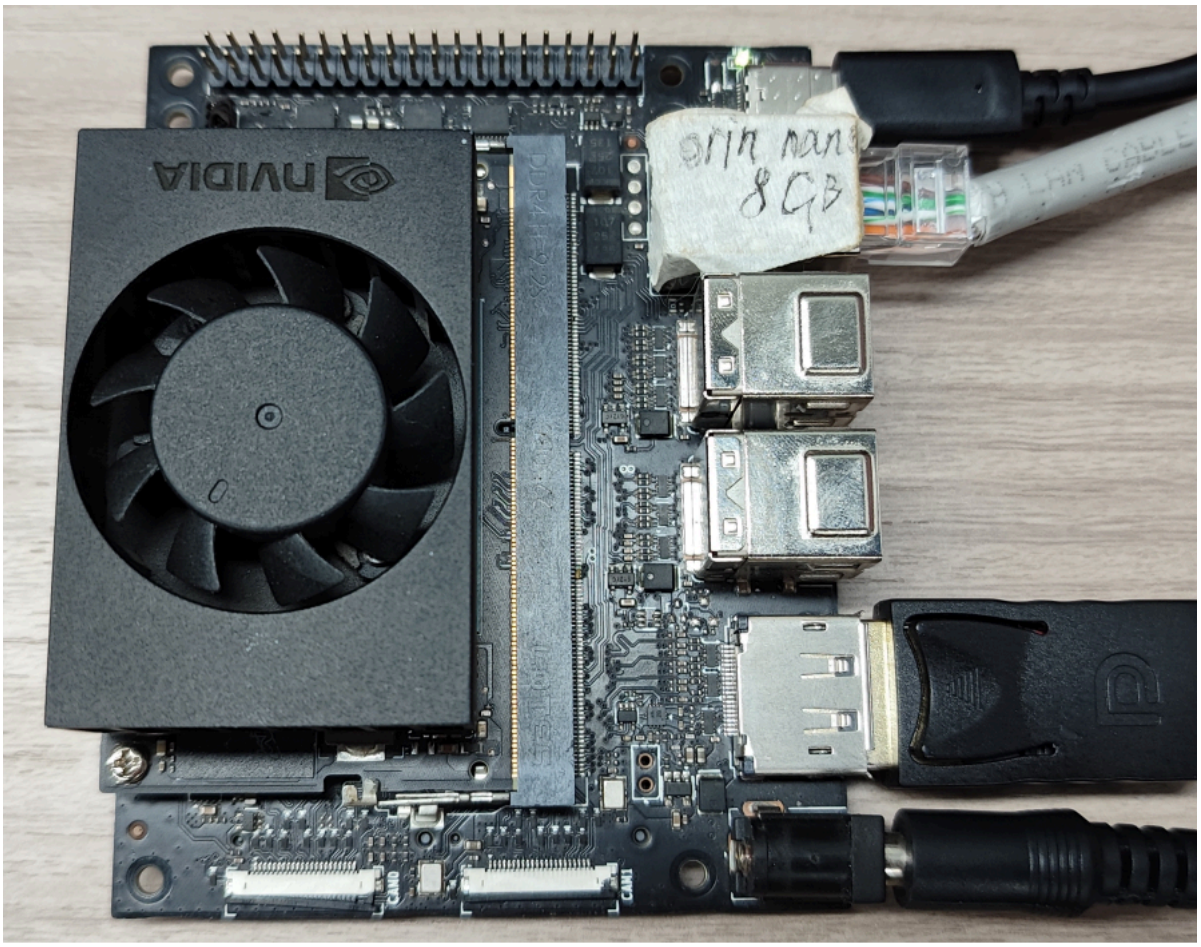
```
The whole process has network requirements, and some components may require a VPN. If the user cannot complete the component installation, use our factory image
```

1. Hardware Connection

The Jetson Orin mainboard needs to be connected to a DC power adapter, a DP data cable, a network cable, and a Type C data cable: the Type C data cable connects to the computer or virtual machine

This illustration is based on the official version of Jetson Orin Nano. Users of other versions can refer to it (the hardware interfaces and functional layout are the same)

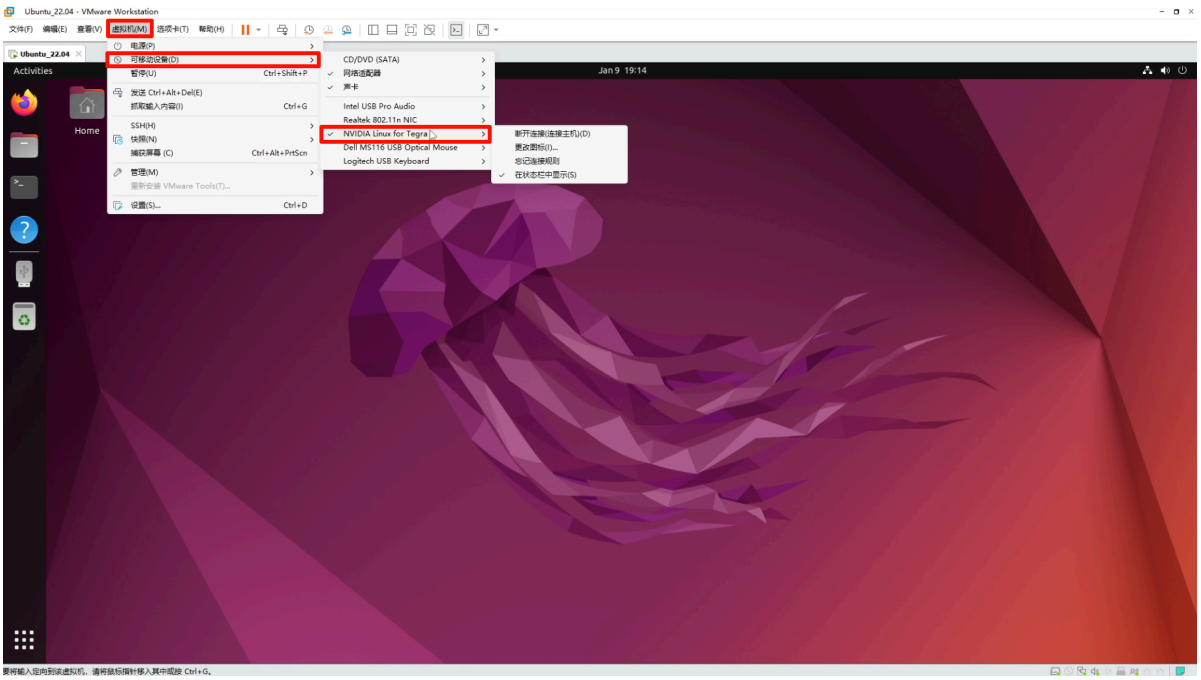
Note: Installing the Jetson environment does not require entering Recovery mode; the mainboard just needs to be powered on!!!



2. Software Connection

The tutorial uses VMware Workstation 17 Pro to write the system!

After starting the virtual machine, select in the menu bar: Virtual Machine → NVIDIA Linux for Tegra → Confirm the connection status with the virtual machine



3. Using SDK Manager

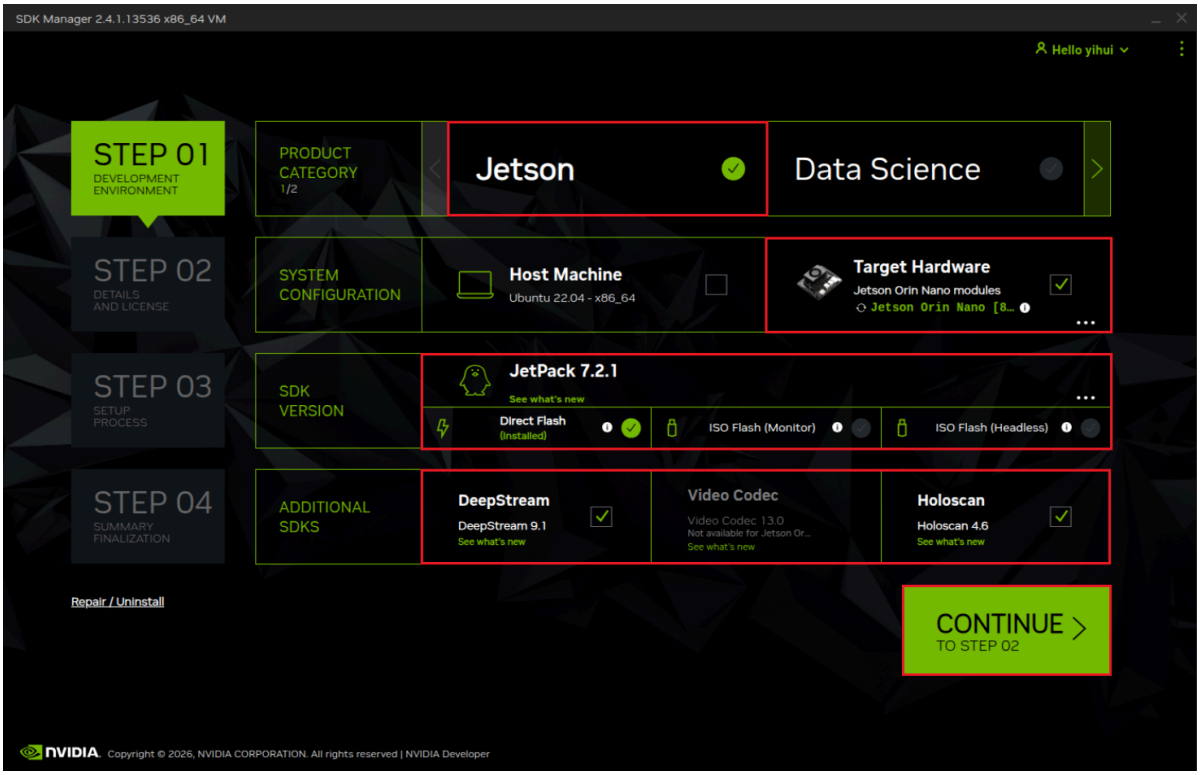
The Jetson Orin mainboard is in the normal system boot state; connect it to the computer or virtual machine with a Type-C data cable.

3.1. Mainboard Selection

After opening `SDK Manager`, select according to your own mainboard model: for the official kit, select `Jetson Orin Nano [*GB developer kit version]`

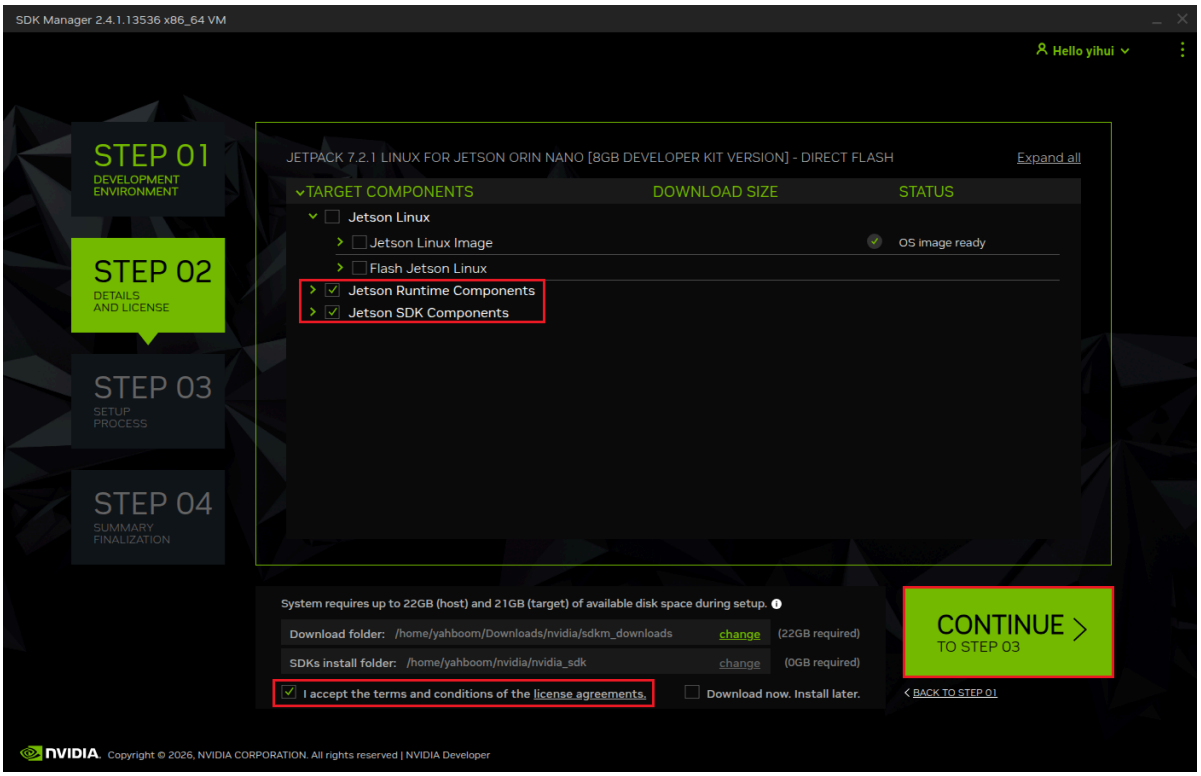
3.2. STEP1

Check the options below, and click `CONTINUE` after confirming that the selections are consistent:

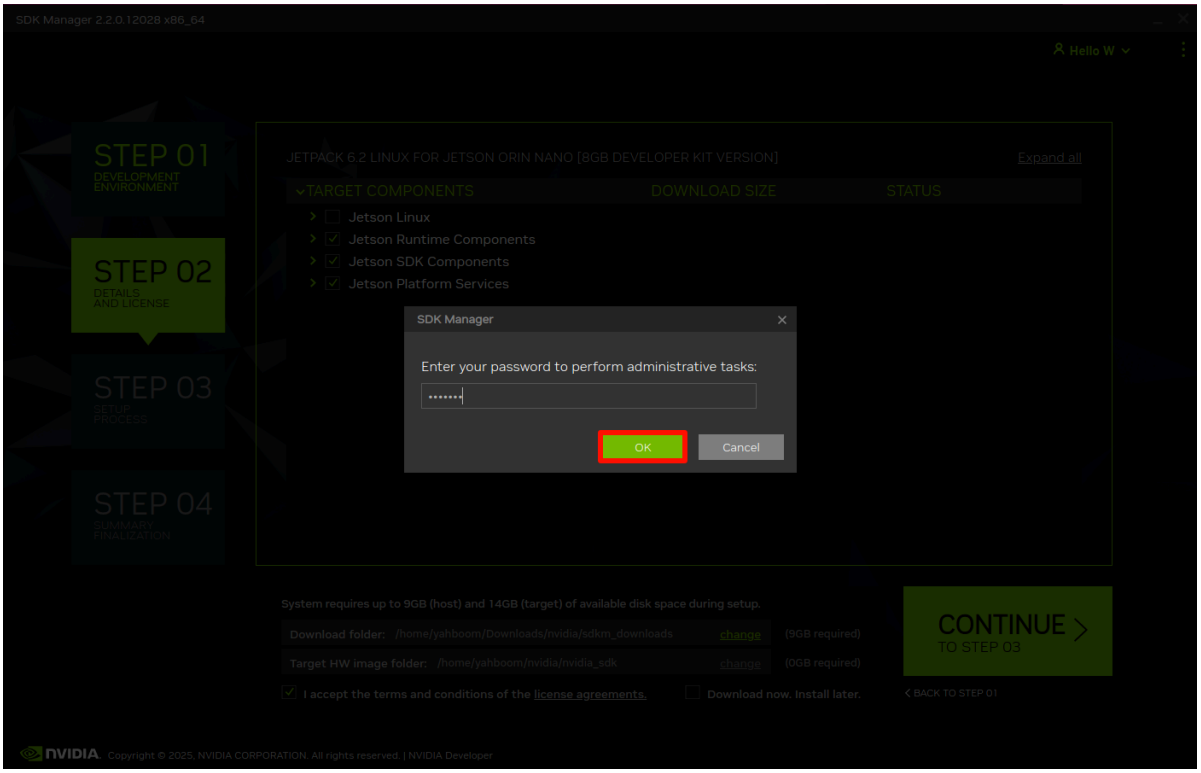


3.3. STEP2

The system is already installed successfully, so only the required component environment needs to be checked: SDK Manager downloads all components automatically, and the next step can continue after the download is complete

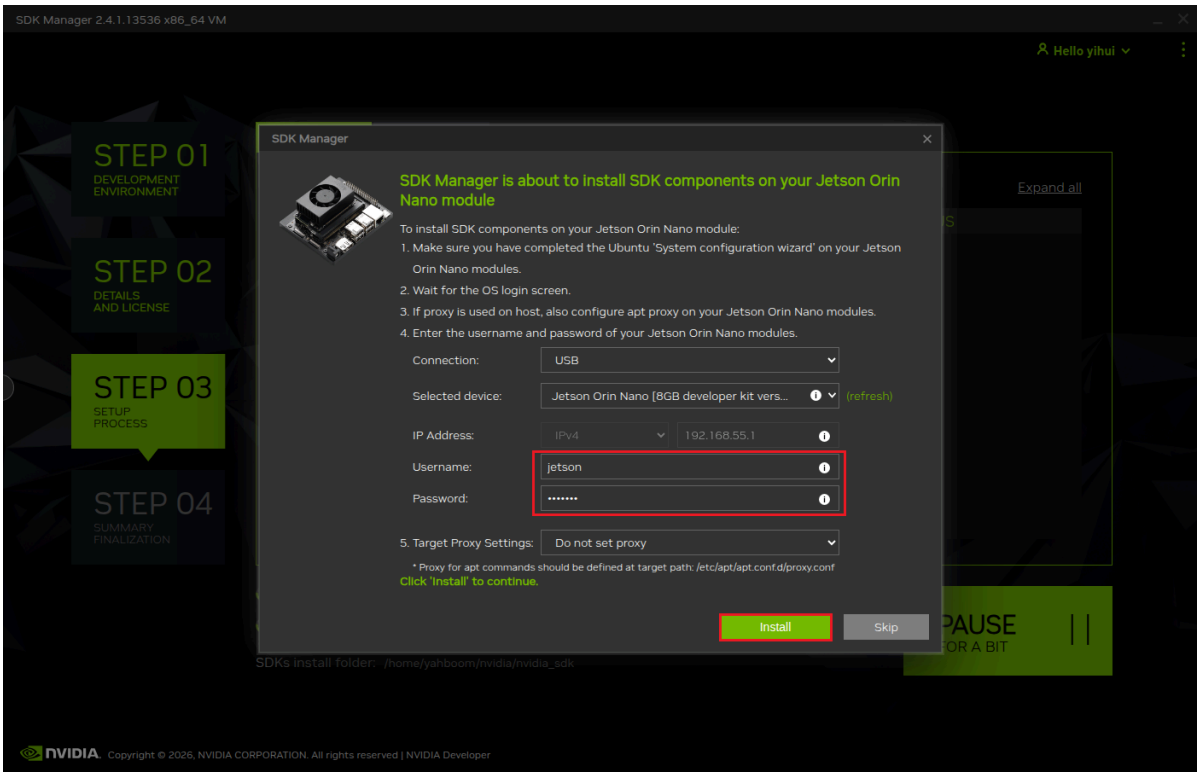


Enter the virtual machine password: yahboom. If it is your own virtual machine, enter your own password

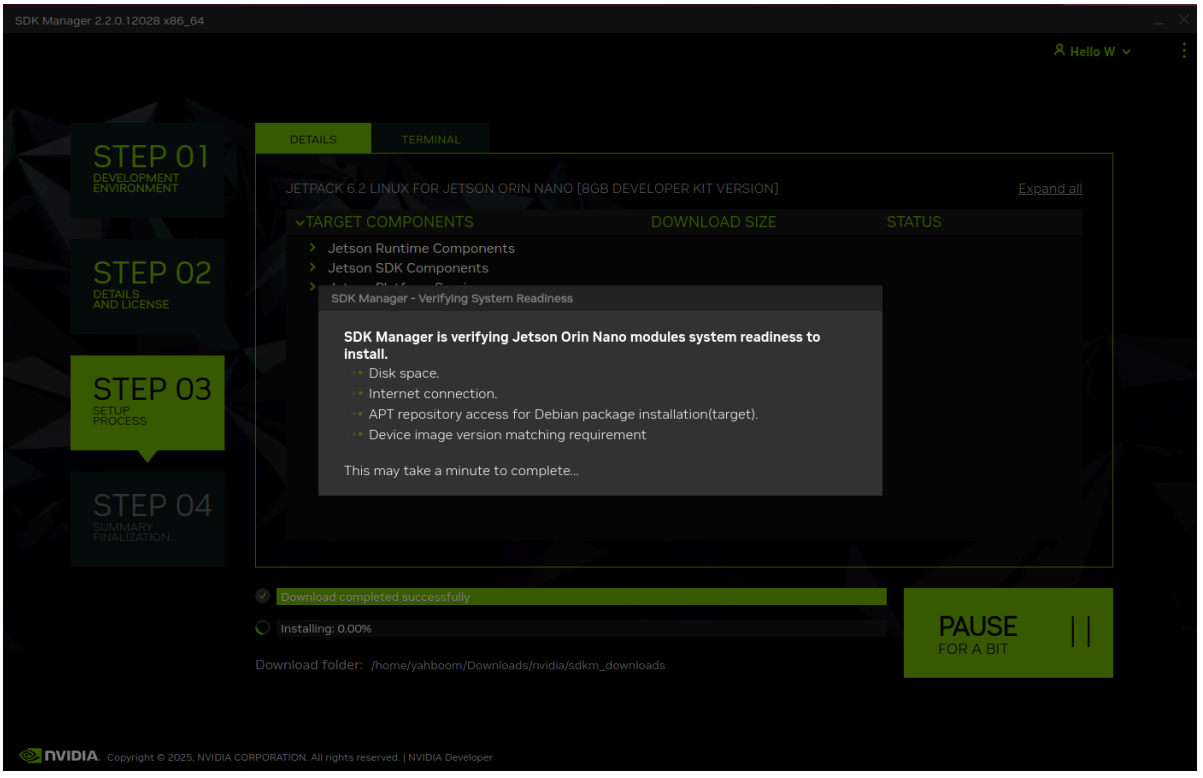


3.4. STEP3

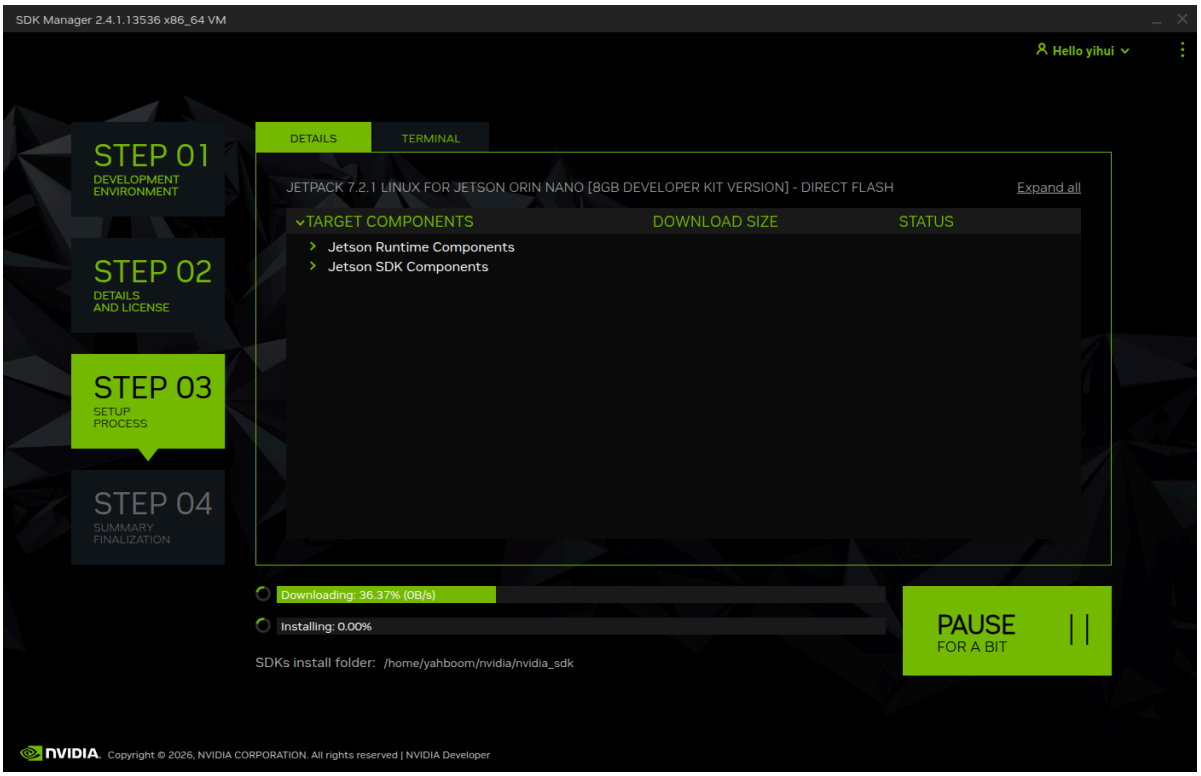
Enter the username and password that were set for yourself when starting the system earlier: the whole process takes a relatively long time, please wait patiently

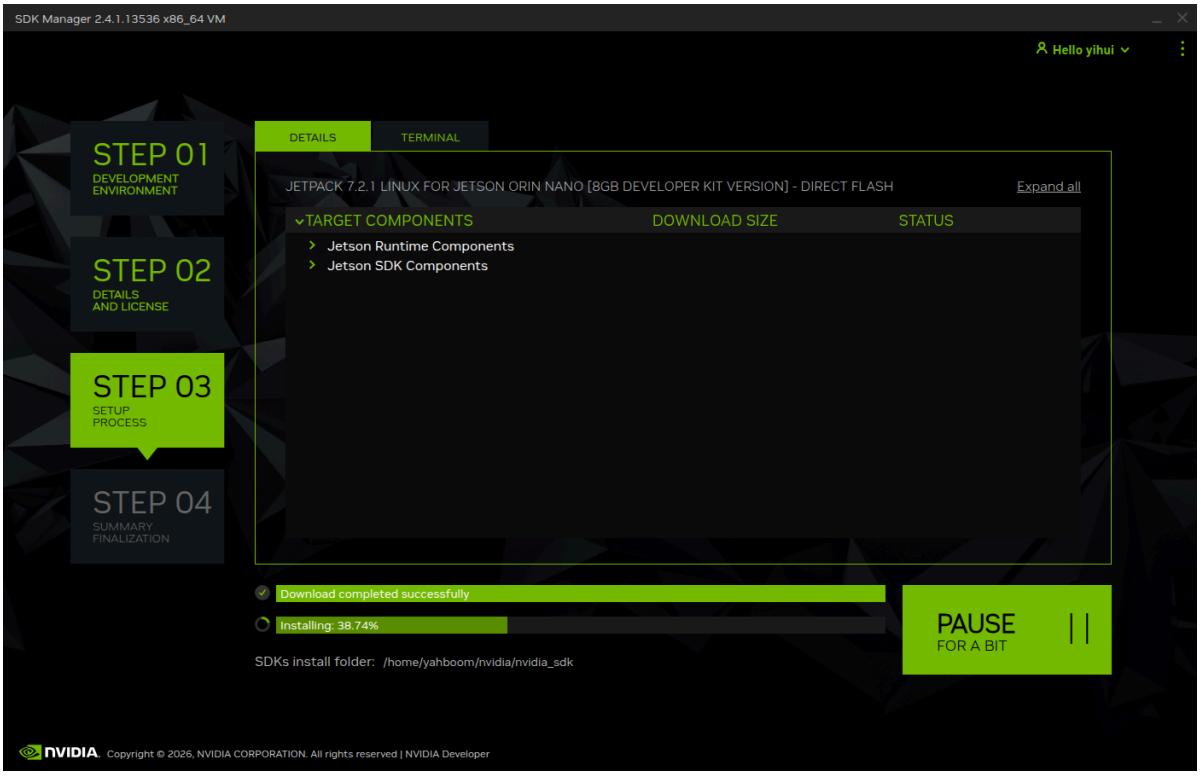


Wait for the system to download and install: if a network failure appears in the virtual machine during the process, it can be ignored. In actual tests it does not affect writing the component environment

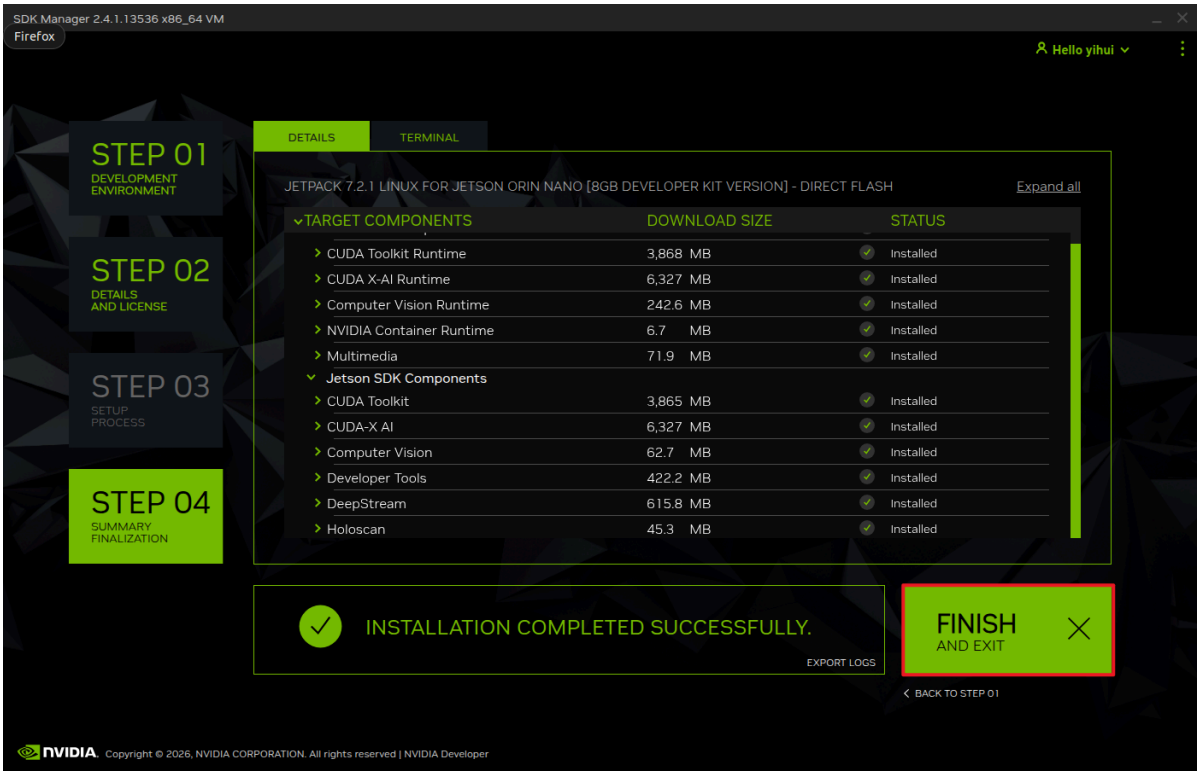


The whole installation process takes a long time, please wait patiently. If the installation fails, reboot the mainboard system and restart SDK Manager to install:


SDK Manager 2.4.1.13536 x86_64 VM
Hello yihui
STEP 01
DEVELOPMENT ENVIRONMENT
STEP 02
DETAILS AND LICENSE
STEP 03
SETUP PROCESS
STEP 04
SUMMARY FINALIZATION
DETAILS
TERMINAL
JETPACK 7.2.1 LINUX FOR JETSON ORIN NANO [8GB DEVELOPER KIT VERSION] - DIRECT FLASH
Expand all
TARGET COMPONENTS
DOWNLOAD SIZE
STATUS
Jetson Runtime Components
Jetson SDK Components
Downloading: 36.37% (0B/s)
Installing: 0.00%
SDKs install folder: /home/yahboom/nvidia/nvidia_sdk
PAUSE
FOR A BIT
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SDK Manager 2.4.1.13536 x86_64 VM
Hello yihui
STEP 01
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JETPACK 7.2.1 LINUX FOR JETSON ORIN NANO [8GB DEVELOPER KIT VERSION] - DIRECT FLASH
Expand all
TARGET COMPONENTS
DOWNLOAD SIZE
STATUS
Jetson Runtime Components
Jetson SDK Components
Download completed successfully
Installing: 38.74%
SDKs install folder: /home/yahboom/nvidia/nvidia_sdk
PAUSE
FOR A BIT
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3.5. STEP4



4. Environment Verification

> **Jtop Tool**

Use the commands below to install Jtop:

```
sudo apt update
sudo apt install python3-pip -y
sudo pip3 install -U jetson-stats --break-system-packages
sudo jtop --install-service
```

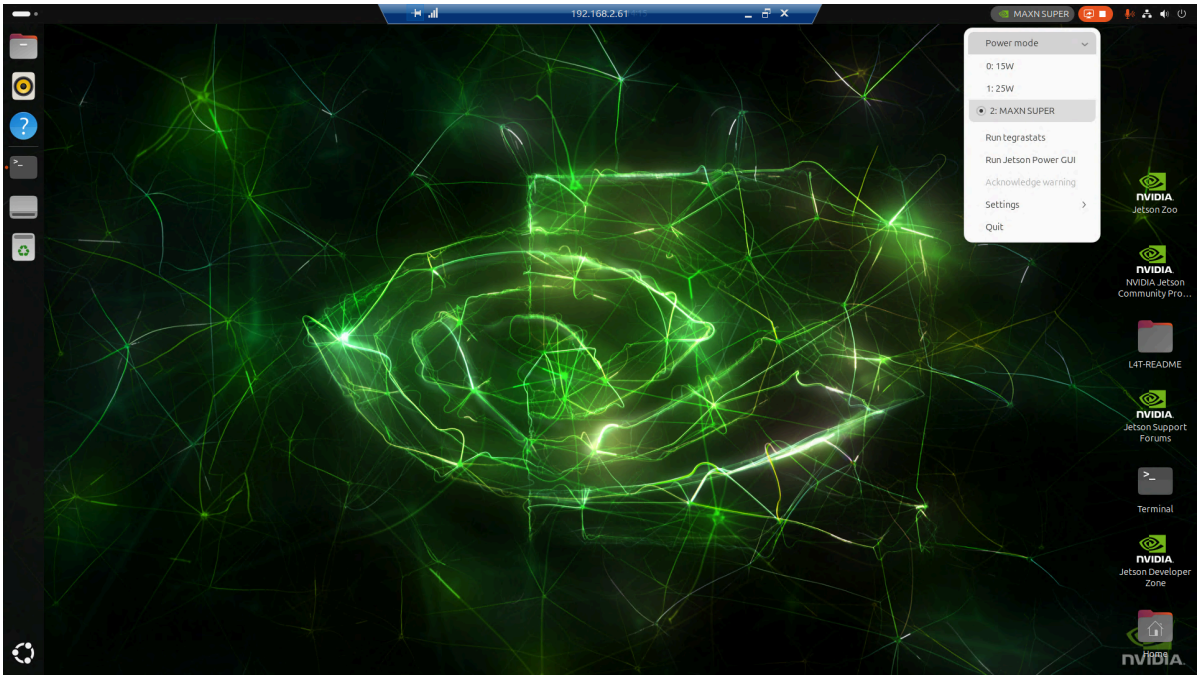
```
jetson@jetson:~$ sudo apt update
[sudo] password for jetson:
Ign:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://ports.ubuntu.com/ubuntu-ports noble InRelease
Hit:4 http://ports.ubuntu.com/ubuntu-ports noble-updates InRelease
Hit:3 https://repo.download.nvidia.cn/jetson/common r39.2 InRelease
Hit:5 https://repo.download.nvidia.cn/jetson/som r39.2 InRelease
Hit:6 http://ports.ubuntu.com/ubuntu-ports noble-backports InRelease
Hit:7 https://apt.kitware.com/ubuntu noble InRelease
Hit:8 https://repo.download.nvidia.cn/jetson/ffmpeg r39.2 InRelease
Hit:9 http://ports.ubuntu.com/ubuntu-ports noble-security InRelease
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
358 packages can be upgraded. Run 'apt list --upgradable' to see them.
jetson@jetson:~$ sudo apt install python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-pip is already the newest version (24.0+dfsg-1ubuntu1.3).
The following package was automatically installed and is no longer required:
  libhash0
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 358 not upgraded.
jetson@jetson:~$ sudo pip3 install -U jetson-stats --break-system-packages
Collecting jetson-stats
  Downloading jetson_stats-7.2.2-py3-none-any.whl.metadata (8.2 kB)
Collecting smbus2 (from jetson-stats)
  Downloading smbus2-0.6.1-py2.py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: distro in /usr/lib/python3/dist-packages (from jetson-stats) (1.9.0)
Collecting nvidia-ml-py>=13.595.45 (from jetson-stats)
  Downloading nvidia_ml_py-13.615.71-py3-none-any.whl.metadata (9.7 kB)
Downloading jetson_stats-7.2.2-py3-none-any.whl (208 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 208.0/208.0 kB 1.4 MB/s eta 0:00:00
Downloading nvidia_ml_py-13.615.71-py3-none-any.whl (58 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 58.1/58.1 kB 7.3 MB/s eta 0:00:00
Downloading smbus2-0.6.1-py2.py3-none-any.whl (11 kB)
Installing collected packages: smbus2, nvidia-ml-py, jetson-stats
Successfully installed jetson-stats-7.2.2 nvidia-ml-py-13.615.71 smbus2-0.6.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with
the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/
warnings/venv
jetson@jetson:~$
```

Note: The system needs to be restarted after the installation is complete!

> **Best Performance Mode**

Enable MAX power mode

```
sudo nvpmodel -m 2 # Jetson Orin Nano
```



Enable Jetson clocks: the CPU and GPU cores all run at maximum frequency

```
sudo jetson_clocks
```

Use the Jtop tool to view system information:

```
jtop
```

